From Manufacturer to Bedside:

Tracing the Future of Medication Management Across the Supply Chain

Building the Intelligent Medication Ecosystem: real-time technologies transforming safety, efficiency, and enterprise value in pharmacy operations.

Table of Contents

The Current State of Medication Management **1**

1. Biopharmaceutical Manufacturing and Distribution **2**

a. Key Considerations for Manufacturers and Distributors 2

b. Technologies to Watch: Radio Frequency Identification (RFID) Tagging 3

c. The Future 3

i. More Granular Medication Tracking 3

ii. Enhanced Visibility and Communication with Partner Facilities 3

2. Hospital Pharmacy **4**

a. Key Considerations for Hospital Pharmacy 4

b. The Future 6

i. Leveraging Smart Systems and Analytics to Optimize Pharmacies 6

ii. Anticipating Regulatory Shifts 6

c. Technologies to Watch: Pharmacy Automation & Centralized Medication Management Software 6

3. Point of Care **7**

a. Key Considerations for Point of Care 7

b. The Future 8

i. Centralized Integration with EHR and Billing Systems 8

ii. Incorporating Technologies that can Limit Errors 8

c. Technologies to Watch: Smart Medication Workstations with Integrated Labelers 9

Building a Safer, Smarter Medication Pathway **10**

The Current State of Medication Management

The path medications take from manufacturer to point of care is complex and often spans the globe. It's also subject to any number of shortages, delays, and diversions. Any weak points in this path are a liability that could negatively impact stakeholders and potentially compromise patient care. Meanwhile, hospital systems and the technologies that power them are changing. New advancements are paving the way for optimizations and improvements in how hospitals manage medications while revealing how many hospital systems are unprepared for the changing tides.

While medical systems often face pushback and challenges when veering from the status quo, regulators are increasing medication management requirements, and the industry is at a crossroads. Further delays to technology updates could put hospitals at a significant and costly disadvantage. To cut through the noise associated with emerging technologies in this space, this whitepaper traces the path of medications from manufacturers to patients. Along the way, we highlight the key challenges relevant to each step and the technologies that we believe are most critical to enabling the next generation of medication management.

1

Biopharmaceutical Manufacturing and Distribution

There is no single starting point for the medication journey because manufacturing spans the globe. The U.S. Food & Drug Administration (FDA) oversees over 23,000 prescription drug products approved for marketing in the United States. Over three-quarters of the active pharmaceutical ingredient (API) manufacturers are located outside the U.S. For biologic drugs, 5,975 different facilities are registered as part of their manufacturing.[1] This is just a small window into the complexity of modern biopharmaceutical manufacturing and the global distribution chain.

Along with guaranteeing the quality of their products, manufacturers are responsible for ensuring medications emerge from their facilities with the labeling and information needed to meet ongoing product identity, quality, safety, and traceability requirements. Conversely, distributors move medications across states, countries, and oceans. Much like manufacturers, distributors are required to maintain thorough records of all their medications. They are responsible for the chain of custody. Any lapse in this system can result in unsafe or compromised medications arriving at hospitals, putting patients in direct danger.

Key Considerations for Manufacturers and Distributors

Managing Supply Chain Disruptions While Maintaining Transparency

Throughout the COVID-19 pandemic supply shortages, the 2021 Suez Canal obstruction, the Francis Scott Key bridge collapse in Baltimore, MD, and ongoing international shipping crises due to conflict in Eastern Europe and the Red Sea, patients have still needed their medications. **These events have illuminated a new reality: medical supply chains will likely continue to face disruptions beyond manufacturer and distributor control**.

The supply chain is a continuous feedback loop across the entire medication lifecycle. The relationship between manufacturers, distributors, and pharmacies is often built on the principle of "trust but verify." Thus, supply chain shortages and disruptions are an ongoing concern for manufacturers, who are often seen as responsible for "fixing" shortages by increasing supply or overcoming disruptions. Along with optimizing processes to support operations when disruptions occur, it's also crucial for manufacturers and distributors to maintain visibility into the movement of medications all the way to the end-pharmacy. These records are particularly important in the event of a recall when all impacted batches need to be located and removed from patient care areas immediately.

Complying with Evolving Regulations

Several regulations enforced by the FDA, namely current good manufacturing practices (CGMPs), require manufacturers to follow strict standards when manufacturing, packaging, and labeling medications. More recently, the Drug Supply Chain Security Act (DSCSA) has placed additional expectations on how medications are labeled and tracked.[2] The regulatory landscape manufacturers and distributors must navigate becomes increasingly complicated with controlled substances due to specific requirements under the Controlled Substances Act (CSA) and additional enforcement by the Drug Enforcement Agency (DEA) regarding licensing, production quotas, security standards, and tracking.

Together, these regulations ensure medications are safe and traceable once they enter the market. They also help ensure pharmacists ordering from various manufacturers can feel confident in the purity and quality of medications when they receive them. However, complying with this breadth of regulations complicates processes for manufacturers and distributors.

Technologies to Watch

Radio Frequency Identification (RFID) Tagging

Sometimes called "the barcode of the future," RFID tags use small radio frequency identification devices to track and identify products. Unlike barcodes, which must be manually scanned, RFID tags don't have to come into direct line-of-sight with sensors, allowing tags to be counted and monitored with minimal effort. As RFID technology continues to improve, it could also unlock additional security and safety by monitoring for changes in temperature or tampering.

The Future

More Granular Medication Tracking

The future revolves around increased transparency in medication tracking. With the implementation of advanced DSCSA requirements, manufacturers are already required to include labels on products that make a wealth of information about the medication available with nothing but a barcode scan. **We anticipate this technology will be the first step of many toward complete supply chain visibility, where suppliers can track medications down to the unit level in real time**.

With this backdrop, we're beginning to see more manufacturers adapting their medication labeling practices. This includes upgrading from traditional barcodes to tags that include RFID. This shift to RFID tagging at the source revolutionizes end-to-end electronic tracking while potentially cutting down on the manual labor associated with scanning medications and notating chains of custody. RFID tags also have much greater data capacity than 2D barcodes and are harder to counterfeit. Thus, this shift will facilitate compliance with the DSCSA regarding labeling while providing additional protection against drug counterfeiting and diversion.

Enhanced Visibility and Communication with Partner Facilities

Software is central to enabling better medication visibility and tracking. We expect to see a rise in the adoption of more advanced software and systems that would allow communication across the medication pipeline. For manufacturers and distributors, this could take the form of software that supports proactive responses to shifts in demand.

Central to this evolution is the concept of interoperability, or the ability of systems to "speak" to each other, and advanced analytics that maximize the data being fed into the system to support decision-making.

RFID Tagging at the Source

Many manufacturers still rely on barcodes, and RFID tags are placed retroactively when a medication arrives at a pharmacy. **RFID tagging at the source, however, can vastly increase visibility into the medication pipeline while simultaneously making medication tracking easier and more robust**.

Benefits of Tagging at the Source:

- Unlocks visibility into medication status and location
- Reduces manual processes throughout the medication journey
- Holds additional information, such as temperature history
- Makes counterfeiting and tampering more challenging

> "The average patient assumes that everyone—every pharmacy and every distributor—is doing everything they can to keep them safe… Except, imagine if we didn't have the data we needed to tell them if a medication got recalled."
>
> **Jon Brown, PharmD, MHA, DPLA**
> Director of Pharmacy
> Indiana University Health

2

The Hospital Pharmacy

The arrival of medications at the pharmacy is the next major step in the medication journey. While many concerns for manufacturers and distributors persist as medications move through pharmacies, such as maintaining their integrity, pharmacy staff must be cognizant of some distinct considerations when moving medications forward.

Key Considerations for Hospital Pharmacies

Maintaining Transparency into Medication Stock and Location

Pharmacies don't just take ownership of medications when they are delivered at the hospital loading docks. They are also largely responsible for moving medications between hospital facilities and departments and ensuring crash carts and medication cabinets are adequately stocked. For large hospital systems, this can be millions of doses of medications being managed within a year. **According to a 2021 CDC survey, 361.3 million drugs were given or prescribed in hospital emergency departments alone**.[3]

Pharmacy managers rely on clear documentation and visibility into the chain of custody to monitor medication integrity, order additional stock, bill accurately, and ensure the security of controlled substances. In addition to the physical location of medications, pharmacies must maintain visibility into their medications' details, such as expiration dates, amounts available, and dosages. Recent regulatory shifts, such as the DSCSA, solidify expectations for pharmacies to possess increased visibility into the status and location of their medications. **However, this transparency can be a major challenge, particularly when working within a larger hospital system where the chain of custody can be complex**.

Managing Pharmacy Inventory and Avoiding Shortages

Ensuring the right medications are available when providers need them is at the core of a hospital pharmacy's mission. In a perfect world, pharmacies would have just enough medication to avoid running out, but not so much that the medication is at risk of expiring before it can be used. Thus, hospital pharmacy staff must walk a tightrope, avoiding running out of therapies in the face of widespread shortages and supply chain disruptions while monitoring for recalls and removing expired or recalled medications from circulation. **In a 2022 survey by the American Society of Health-System Pharmacists, 99.7% of pharmacists reported being affected by shortages of critical sterile injectables, and 7% reported at least one shortage-related medication safety event that caused some degree of patient harm**.[4] To manage these shortages, pharmacists reported taking actions like switching to therapeutic alternatives, converting from intravenous to oral dosage forms, or changing clinical order sets or protocols — all of which are changes that directly impact the day-to-day operations of pharmacy staff.

Optimizing pharmacy inventory is essential to ensuring the "five rights" of care — this means making sure the right medication is given to the right patient, at the right dose, via the right route, and at the right time. It is also important for the hospital's bottom line. Hospitals are often required to pay the cost of expired or unused medications, increasing the pressure on pharmacists to avoid over-ordering. **In 2023, drug expenditures were $37.1 billion in nonfederal hospitals and $135.7 billion in clinics, and those numbers are expected to increase**.[5] If not managed intelligently, medication waste represents not only a major financial drain —but a missed opportunity to drive safety, efficiency, and enterprise value across the entire medication ecosystem.

Optimizing Pharmacy Processes

Ensuring safe, effective, and efficient medication use for optimal patient outcomes is the number one goal for pharmacy staff. Stocking or medication management errors can directly impact the pharmacy's ability to serve patients. On the other hand, there is ongoing pressure to operate more efficiently. Hospital pharmacy staff can find themselves caught in a catch-22 as a result. Moving quickly to optimize their time can result in human errors that put patients at risk. Slowing down, however, can cost workers time, translating to lost money. In parallel with the pressure to work faster is the pressure to work more. **In a 2021 survey, 97% of pharmacy administrators reported increasing the use of overtime to fill shifts, and 89% reported using pharmacists to fill pharmacy technician shifts or perform technician duties**.[6] Pharmacy staff must have the tools they need to work smarter while navigating a complex and dynamic environment.

Data Centralization and Interoperability

Historically, hospital systems have managed various disconnected systems, each with a different function and goal. The result was a system where data was collected and then kept siloed. This approach limits visibility into what is truly happening with inventory and reduces system-wide efficiency. Today, technology makes it possible to link systems to create a connected environment where data is exchanged and organized.

Two essential functions are necessary to make this work:

- **Interoperability**, or the ability of systems to "speak" to each other.
- **Data centralization**, or the strategy that determines how and when data should flow between systems and enables the creation of a unified data repository.

Together with more granular medication tracking, interoperable smart systems can help generate meaningful and actionable insights. Specifically, this integration can give hospital pharmacy systems the framework required to establish one source of truth for medication management and the insights needed to replace time-consuming, manual steps with automated workflows.

> "Having [medications] RFID tagged so that the central pharmacy and others can go ahead and look at an inventory that's live and in real time makes it so that there is much less of a need for the practitioner to perform some of those tasks."
>
> **Daniel Ness, MD**
> Associate Medical Director
> Rady Children's Hospital - San Diego

The DSCSA and Hospitals

The DSCSA, passed by the FDA in 2013, is regarded by many as an opportunity to improve patient safety. The regulation enables traceability by specifying that pharmaceutical products must be marked with four data elements—a national drug code (NDC), serial number, lot number, and expiration date. While the DSCSA allows for this information to be stored using two-dimensional (2D) barcodes (e.g., GS1 DataMatrix barcode), it also allows the use of more advanced labels that utilize RFID instead.

Driven by DSCSA requirements, pharmaceuticals in packages or homogeneous cases can be tracked and traced from manufacturers to the pharmacy. Yet, the management of medicines throughout hospitals—from receipt to administration—is not within the scope of DSCSA.

While medication management is not required at the unit level within hospitals, many anticipate that the future of medication management and the DSCSA will require full supply chain visibility.

The Future

Leveraging Smart Systems and Analytics to Optimize Pharmacies

Smart systems that increase visibility without increasing labor can address safety and efficiency challenges.

We anticipate a rise in the number of hospitals using systems that make it quick and easy to count medication inventory, track their location within the larger hospital network, and enable more comprehensive insights into inventory levels. Many of these systems use a combination of hardware (such as RFID tags and smart medication dispensing systems) and software with built-in analytics that can maximize the use of the available data.

With this technology, hospital decision-makers can have accurate, synchronized data from a variety of sources, such as automatic dispensing cabinets and narcotic vaults, in one place. Applying advanced analytics to this data can then provide insights on ordering, utilization, and future use forecasts. Ultimately, this information can reveal opportunities for medication waste reduction and facilitate improved billing practices, such as more accurate charge capture, both of which have financial benefits.

Anticipating Regulatory and Industry Shifts

As technology advances, hospitals face mounting pressure to modernize and optimize their pharmacy workflows and management. In addition to pressure from regulators to enable more effective communication between stakeholders in the medication pipeline, there is a growing movement in favor of certain new technologies, like artificial intelligence (AI). **While it might still be in the relatively early stages of adoption, AI is increasingly common in various hospital settings and has the potential to revolutionize the use of medication management data, driving new insights and actions as a result**.

Technologies to Watch

Pharmacy Automation Systems

Pharmacy systems that use technology, such as RFID, to simplify medication counting can cut down workflow steps and potentially scan hundreds of medications in seconds. These systems have the potential to track medications with a high degree of accuracy with less time and energy.

Centralized Medication Management Software

As its name implies, centralized medication management software helps pharmacies monitor medications across an entire hospital system in one central place. This helps pharmacies integrate data from a wide range of sources and can improve visibility into the entire medication supply chain.

> "For me, it always comes back to 'let's make it easy to do the most correct thing'. Does it take some planning, budgeting, and resources? Absolutely. But five years from now, that technology is going to be the standard."
>
> **Jon Brown, PharmD, MHA, DPLA**
> Director of Pharmacy
> Indiana University Health

3

Point of Care

After a long journey from manufacturer to pharmacy, the point of care is the final stop for medications. Point-of-care medication management can look different for different hospital departments, but each care provider is united by a common purpose—to administer medications to patients safely.

Key Considerations at the Point of Care

Accurate Medication Administration

Ensuring the medication being given to a patient is correct and that it is not counterfeited, expired, or recalled is essential to medication administration. In reality, however, care providers might juggle multiple manual steps in a hectic environment when administering medications, leaving room for human error. **Some studies estimate that more than 200,000 patient deaths annually are due to preventable medication errors**.[7]

Facilitating Accurate Record-Keeping

In addition to administering medications, care providers are responsible for documenting their use—potentially in multiple places, such as the patient's electronic health records (EHR) and the hospital billing system. If a controlled substance is being administered, additional records are required, along with follow-up disposal steps when medication is left over. It is essential for hospitals to possess an accurate and transparent chain of custody to adhere to 340B compliance, which affects reimbursement, ensures accountability at every hand-off, and clearly states which patients received which medications.[8] While these records are essential to the hospital's operation, they represent multiple hurdles that can impede workflows, increase burnout, and hamper the patient-care provider relationship.

Patient Safety and Anesthesiology

Anesthesia providers are often in an operating room environment in which medications must be administered in seconds. While anesthesia providers aren't technically hired to monitor operating room (OR) medication inventory, they often double as medication managers. And when medications are unavailable, it isn't out of the question for surgical staff to cancel or delay surgery.[9]

Even if providers have the medications they need, the functionality of the medication dispensing systems can impede their work. A clunky workflow that makes it hard for anesthesiologists or anesthesiology techs to get to medications quickly can cost precious time. In some cases, anesthesiologists might carry medications with them or pull them from trays ahead of time to avoid being unable to get to a medication.

In this environment, it is more critical than ever that providers have an intuitive process that ensures they have constant visibility into stock, quick access to the right medications, and streamlined administration recording processes.

> "[Medications] need to be accessible quickly because often the time from the decision that you need the medication to the time that you want to have it in the patient is seconds to a minute. So having custody of medications such that you know where they are and that they're available to you with minimal steps to get access is really important."
>
> **Daniel Ness, MD**
> Associate Medical Director
> Rady Children's Hospital - San Diego

Uncovering Drug Diversion

Drug diversion, or the intentional removal of medications from the healthcare system for unlawful distribution or use, is an unfortunate reality. While diversion can happen at any point in the medication journey, the point of care is a particularly vulnerable moment for some of the most frequently diverted medications, such as opioids. In fact, experts believe the ongoing opioid crisis in the U.S. is a major driver of diversion in medical facilities.[10] Healthcare providers (HCPs), including physicians and nurses, are 10-15% more likely to abuse substances, including opioids and other prescription medications, than the general public.[11]

Diversion can have far-reaching impacts on patients, HCPs, and medical systems, including the risk of death from overdose, removal of critical medications, and erosion of trust in medical providers.[12] For patients, there are the additional risks of receiving inadequate care or exposure to unsafe materials, such as contaminated syringes or medications that have been tampered with. In one instance, at least 12 hepatitis C viral infections were identified in patients who received opioid injections from a nurse who admitted to diverting injectable narcotics.[13]

For the same reasons that medication inventory can be hard to track, drug diversion can be even harder to spot. **A study estimated nearly one-third of healthcare institutions investigate 11 or more drug diversion cases each year, yet 4 in 5 healthcare executives believe that most drug diversion goes undetected**.[14] Fortunately, technologies such as AI, medication management analytics, and RFID-tracking labels can help systems establish anti-diversion programs.

The Future

Centralized Integration with EHR and Billing Systems

We anticipate that integrating medication management systems with EHRs and hospital billing systems will be essential in the near future. This level of centralized integration has the potential to remove manual steps from provider workflows while providing visibility into the chain of custody and administration of medications. Instead of retroactively filling in medication information, potentially hours or days after the administration takes place, hospital staff can more seamlessly record all necessary information at the exact moment care takes place.

By helping these disparate systems "talk" to each other, hospital pharmacy managers will be able to acquire data that can help them ensure the right meds are on hand at the right time, whether at the patient's bedside or in an operating room. It can also close the loop on the location of medications, particularly in the event of a recall or suspected diversion event.

Incorporating Technology That Can Limit Errors

Medication errors can be fatal, and many current workflows focus on preventing them. Technology systems enabling providers to function confidently and spend more time on patient care will be essential in the future. **These systems, such as smart medication administration stations, have the potential to facilitate safe medication administration by flagging when medications are expired, presenting clear medication information on a screen, and providing a clear station inventory of what medications are on hand at any given moment**.

Technologies to Watch

Smart Medication Workstations with Integrated Labelers

Workstations are essential to medication administration, and workstations that simplify providers' workflows are becoming increasingly advanced. Some workstations can already integrate with EHRs and include integrated medication labelers that meet Institute for Safe Medication Practices (ISMP) guidelines for safe medication handling.

> "Once we realized you can use RFID information to trace back who had access to vials of a medication during a diversion event, we understood that this technology is a great way of tracking medications. We also saw the added value of extending our tracking beyond the pharmacy and into the operating room."
>
> **Daniel Ness, MD**
> Associate Medical Director
> Rady Children's Hospital - San Diego

Advancing the Medication Pathway: Smarter, Safer, More Connected

Today's medication pathway is more complex—and more critical—than ever before. As the demands on health systems grow, hospitals must evolve beyond fragmented workflows and legacy tools. Modern digital technologies offer the opportunity to transform medication management from a series of disconnected handoffs into a unified, intelligent ecosystem.

At the heart of this transformation is the patient. Safe, effective, and equitable care depends on real-time visibility, proactive decision-making, and seamless coordination across every step of the medication lifecycle. By embracing innovation and elevating standards, healthcare leaders can reduce waste, improve compliance, and, most importantly, safeguard patient outcomes in every moment that matters.

The Intelliguard Smart Path

Experience a unified, cloud-based intelligence platform that delivers real-time medication visibility—from manufacturer to patient.

Explore the Mira Ecosystem™

References

1. U.S. Food and Drug Administration. FDA Regulated Products and Facilities. Published online October 2024. Accessed March 27, 2025. https://www.fda.gov/about-fda/economics-staff/fda-glance
2. U.S. Food and Drug Administration. Drug Supply Chain Security Act (DSCSA). Accessed March 27, 2025. https://www.fda.gov/drugs/drug-supply-chain-integrity/drug-supply-chain-security-act-dscsa
3. National Center for Health Statistics. National Hospital Ambulatory Medical Care Survey. 2021. Accessed March 30, 2025. https://www.cdc.gov/nchs/fastats/drug-use-therapeutic.htm
4. American Society of Hospital-Pharmacists. Severity and Impact of Sterile Injectable Drug Shortages. Published online 2022. Accessed March 31, 2025. https://www.ashp.org/-/media/assets/drug-shortages/docs/ASHP-Sterile-Injectable-Drug-Shortages-March2022.pdf
5. Tichy EM, Hoffman JM, Tadrous M, et al. National trends in prescription drug expenditures and projections for 2024. *Am J Health Syst Pharm*. 2024;81(14):583-598. doi:10.1093/ajhp/zxae105
6. American Society of Hospital-Pharmacists. Pharmacy Technician Shortage Survey Findings. 2022. https://www.ashp.org/-/media/assets/pharmacy-technician/docs/Technician-Shortage-Survey-Exec-Summary.pdf
7. Rodziewicz TL, Houseman B, Vaqar S, Hipskind JE. Medical error reduction and prevention. *In: StatPearls. StatPearls Publishing*; 2025. PMID: 29763131.
8. Health Resources and Services Administration. 340B Drug Pricing Program. 2025. Accessed May 2, 2025. https://www.hrsa.gov/opa
9. Cardinal Health. Survey Finds 40 Percent of Health Care Providers Have Canceled Surgical Cases Due to Lack of Supplies. 2018. Accessed March 27, 2025. https://www.multivu.com/players/English/8286651-cardinal-health-hospital-supply-chain-management-survey/
10. U.S. Centers for Disease Control and Prevention. Clinician Brief: Drug Diversion. March 8, 2024. Accessed April 9, 2025. https://www.cdc.gov/injection-safety/hcp/clinical-overview/index.html
11. American Addiction Centers. Statistics for Substance Use in Medical Professionals. November 21, 2024. Accessed April 9, 2025. https://americanaddictioncenters.org/rehab-guide/addiction-statistics-demographics/medical-professionals
12. Berge KH, Dillon KR, Sikkink KM, Taylor TK, Lanier WL. Diversion of drugs within health care facilities, a multiple-victim crime: patterns of diversion, scope, consequences, detection, and prevention. *Mayo Clin Proc*. 2012;87(7):674-682. doi:10.1016/j.mayocp.2012.03.013
13. Njuguna HN, Stinson D, Montgomery P, et al. Hepatitis C Virus Potentially Transmitted by Opioid Drug Diversion from a Nurse - Washington, August 2017-March 2018. *MMWR Morb Mortal Wkly Rep*. 2019;68(16):374-376. doi:10.15585/mmwr.mm6816a3
14. Wolters Kluwer. The State of Drug Diversion 2023 Report. 2023. Accessed March 31, 2025. https://www.wolterskluwer.com/en/expert-insights/the-state-of-drug-diversion-2023-report